Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

Letterkenny Campus
Port Road, Letterkenny
County Donegal,
F92 FC93, Ireland
T: +353 (0)74 918 6000

Killybegs Campus
Shore Road, Killybegs
County Donegal,
F94 DV52, Ireland
T: +353 (0)74 918 6600

atu.ie

Tesseract OCR Set up

Pre-requisites:

- Anaconda Navigator installed and set up
- Jupyter notebook working
- Windows platform

Tesseract OCR Setup Guide (Windows + Conda + Jupyter + requirements.txt)

- Create a Conda environment
- Install all Python OCR libraries from a requirements.txt file
- Install Tesseract OCR (Windows)
- Configure Jupyter Notebook to use Tesseract
- Run a working OCR test

This guide works on restricted laptops.

Note some important clarification about Tesseract:

- Tesseract is an OCR engine — a program that can read text from images
- It is free and open source and maintained by Google
- It is not a Python package
- It is not a machine-learning library
- It is not installed with pip install pytesseract (this is a python wrapper library)
- It does not depend on NumPy or OpenCV(these libraries are used before OCR to load or preprocess images)

Tesseract is an Optical Character Recognition (OCR) library for Python, which can be used to read and recognise text in images, e.g. of pages, license plates, etc. To use it, we begin by importing the required modules.
There are different versions available, where the latest is v5. Tesseract v5 - the modern LSTM-based (LSTM - Long Short-Term Memory, a type of deep learning network) version, which is a neural network engine making it faster and more accurate than older tesseract versions

Version	Year	OCR Engine	Accuracy	Speed Key	Features
1.x	2006–2007	Legacy	Low	Fast	First open-source release
2.x	2007–2008	Legacy	Low-Med	Fast	More languages, small updates
3.x	2010–2018	Legacy + Cube	Medium	Medium	Layout analysis, hOCR, more languages
4.x	2018–2021	LSTM + Legacy	High	Medium	Deep learning OCR, big accuracy jump
5.x	2021–present	LSTM + Legacy	High	Med- Fast	Modern codebase, best models

Letterkenny Campus
Port Road, Letterkenny
County Donegal,
F92 FC93, Ireland
T: +353 (0)74 918 6000

Killybegs Campus
Shore Road, Killybegs
County Donegal,
F94 DV52, Ireland
T: +353 (0)74 918 6600

atu.ie

Steps

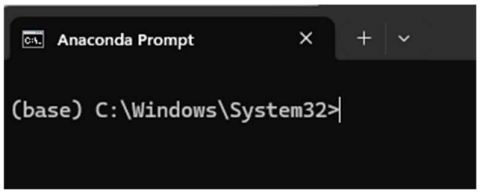
1. In your labs folder create a requirements.txt file named : requirements.txt

Add the following contents :

```
pytesseract
opencv-python
pillow
numpy
matplotlib
easyocr
ipykernel
pymupdf
```

(You can add more libraries later if needed.)

2. Create and activate a Conda environment: Open Anaconda Prompt:



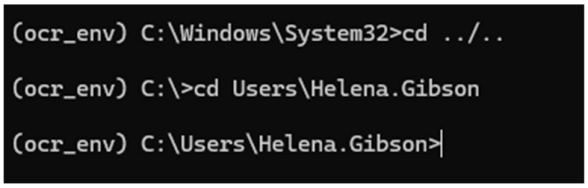
Type :

```
conda create -n ocr_env python=3.10
conda activate ocr_env
```

3. Change directory to where the lab and requirements.txt are located

```
cd <path_name>
```

Example:



4. Install everything from the requirements.txt: Type

```
pip install -r requirements.txt
```

5. Add the conda environment to Jupyter: This adds a kernel that can be selected in your Jupyter notebook. Type

```
python -m ipykernel install --user --name ocr_env --display-name
"Python (ocr_env)"
```

6. Install Tesseract OCR for Windows

Download the official Windows installer file from https://digi.bib.uni-mannheim.de/tesseract/ or (also added BB for convenience)

```
tesseract-ocr-w64-setup-5.5.0.exe (64-bit)
```

Double click the installer exe file to install the full tesseract application
If admin rights are available, install to:

```
C:\Program Files\Tesseract-OCR\
```

If admin rights are ***not*** available, choose a folder inside the user profile (this works and avoids permission issues) e.g.:

```
C:\Users\<yourname>\Tesseract-OCR\
```

Students must know the full path to tesseract.exe, as this will be added to the python notebook.

```
Example: C:\Users\Helena.Gibson\tesseract\tesseract
```

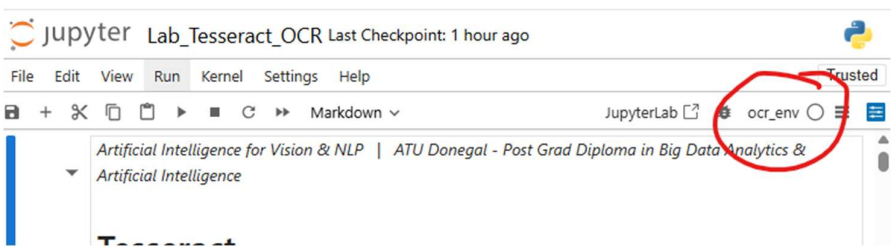
7. In your Jupyter notebook import the python wrapper(pytesseract) and set the location of tesseract application. Jupyter does NOT automatically detect Tesseract.

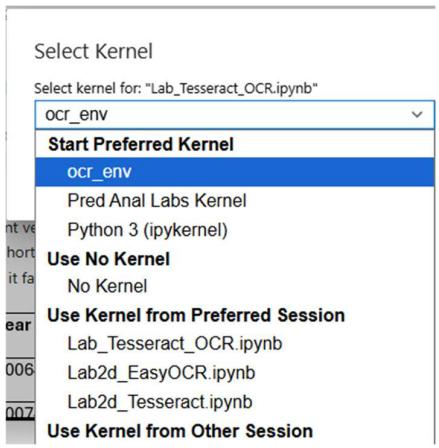
```
pytesseract.pytesseract.tesseract_cmd =
r"C:\Users\<yourname>\Tesseract-OCR\tesseract.exe"
```

Example:

```
import pytesseract
from pytesseract import Output
# change this path name to where you have stored tesseract.exe
pytesseract.pytesseract.tesseract_cmd = r"C:\Users\Helena.Gibson\tesseract\tesseract\tesseract.exe"
```

8. In the Jupyter Notebook the kernel can now be selected to the one just created – ocr_env





9. The tesseract lab python notebook should now execute successfully.